ScreenR 1.15: statistical revision, new infrastructure and a benchmark against MAGeCK

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1 Summary

ScreenR is a Bioconductor package for the analysis of pooled shRNA screens. This report documents a revision of the package (versions 1.15.1 to 1.15.11) and a benchmark of its hit-finding methods against MAGeCK, the most used tool for pooled screens. The revision

- fixed a build failure on Bioconductor and several statistical errors that changed the results (edgeR fitted on normalized instead of raw counts, a barcode filter that was never applied, a wrong design matrix in the vignette);
- replaced the legacy hit rules with tests that return calibrated p-values and false discovery rates (FDR);
- made the ScreenR object a `SummarizedExperiment`, accepted any input format and added quality-control functions, a new real dataset with known hits and five new vignettes.

The benchmark used 30 simulated screens with known hits in six scenarios and a published K562 shRNA essentiality screen with reference essential and non-essential genes. The legacy ScreenR rule called about half false hits; the new methods keep the FDR at or below the nominal level, rank true hits as well as MAGeCK and run in seconds.

2 Background

In a pooled shRNA screen a library of short hairpin RNAs (shRNA, “barcodes”), several per gene, is introduced into cells; after a selection (a drug, or simply time in culture) the abundance of each barcode is measured by sequencing. A gene is a hit when its barcodes are consistently depleted or enriched in the treated cells compared with the controls.

Before this revision ScreenR found hits with three approaches:

1. **Z-score rule** (`find_zscore_hit()`): the log2 fold change of each barcode is converted into a Z-score and a gene is a hit when at least `number_barcode` (default 6) of its barcodes are below a threshold. The rule has no p-value and no error control.
2. **CAMERA** (`find_camera_hit()`): the limma competitive gene set test, with the barcodes of each gene as a set.
3. **ROAST** (`find_roast_hit()`): the limma rotation gene set test.

The hits of the three methods were then intersected.

3 Software changes

3.1 Bioconductor build failure (1.15.1)

The vignette failed on the Bioconductor builders with `BLAS/LAPACK routine 'DPOTRF' gave error code -4`. `find_camera_hit()` called `edgeR::glmLRT()` with all the columns of the design matrix as coefficients, including the intercept, which recent versions of edgeR reject. The likelihood ratio test was not used by CAMERA and was removed.

3.2 Statistical errors that changed the results (1.15.2 – 1.15.6)

Function	Error	Fix
`create_edger_obj()`	edgeR was given CPM-normalized counts, so its library sizes and dispersions were wrong	raw counts, annotation matched by barcode
`find_camera_hit()`, `find_roast_hit()`	the `number_barcode` filter was computed and discarded: every gene was tested	genes need more than `number_barcode` barcodes
vignette	the design `model.matrix(~ groups)` was renamed with `unique(groups)`, whose order differs from the columns: the "Time3 treated" contrast compared Time3 controls with the baseline	`create_design_matrix()`, `create_contrast()`
`compute_metrics()`	robust Z-score not standard; log2 ratio with a 1e-7 offset that exploded for low counts	`(x - median) / mad`; pseudocount 0.5 CPM
`compute_slope()`	regression on the row number instead of the time	regression on the time points
`find_camera_hit()`	`thresh` and `lfc` had no effect	deprecated
`find_time_hit()`	no main effect of the treatment, so a constant effect looked like a growing one	model `a + b·time + d·treated + c·time·treated`
examples	treatment and control swapped in `compute_metrics()`	corrected

An independent review (a second agent with no access to the development history) re-checked every change of 1.15.4; its findings are the 1.15.6 fixes.

3.3 New statistical functions (1.15.4)

- `create_design_matrix()` and `create_contrast()` build designs and contrasts by group name; contrasts can be names, vectors or `makeContrasts()` matrices.
- CAMERA and ROAST gained an FDR cut off and a gene log2 fold change.
- `normalize_data()` gained TMM (edgeR) and control-barcode normalization.
- `compute_barcode_statistics()`: replicate-aware barcode statistics from the edgeR quasi-likelihood F-test.
- `find_gene_zscore_hit()`: gene-level test with FDR for the Z-score approach (reworked in 1.15.10, see below).
- `find_time_hit()`: one model for the whole time course.
- `summarise_hits()`: combines the methods with a per-gene intersection-union test (MaxFDR).

- Quality control: `compute_gini()`, `plot_gini()`, `plot_replicate_correlation()`, `find_discordant_barcodes()`.

3.4 More general input (1.15.5)

`create_screenr_object()` accepts any column names (`barcode_column`, `gene_column`), describes samples with a `sample_table` when their names do not follow the `Day_Treatment_Replicate` convention, and checks the input (groups, non-numeric columns, duplicated or unannotated barcodes) with messages that suggest the right column or argument.

3.5 ScreenR object as a SummarizedExperiment (1.15.8)

`screenr_object` now extends `SummarizedExperiment`: raw and normalized counts are assays, the barcode annotation is `rowData()` and the sample description `colData()`. The object works with Bioconductor tools and subsets consistently (`object[barcodes, samples]`). Before the change, every exported function was run on the example data and its output saved; after the change all 100 outputs were identical. Objects saved with older versions are converted automatically (`update_screenr_object()`).

3.6 K562 dataset (1.15.9)

To evaluate methods on real data with known hits, ScreenR now ships a subset of the genome-wide K562 shRNA essentiality screen of Morgens et al. (2016): plasmid library vs two replicates after 14 days of growth, with the core essential (CEGv2, 681 genes, true hits) and non-essential (NEGv1, 904 genes, true negatives) reference genes of Hart et al. The subset (67,358 shRNAs: all the reference genes, 1000 random other genes and 1000 non-targeting shRNAs) keeps the package under the Bioconductor size limit. The data are BSD-2 (casTLE) and MIT (BAGEL) licensed; the build script is in `inst/scripts`.

3.7 Documentation

Five new vignettes: “Improved statistical analysis with ScreenR”, three step-by-step tutorials (creating the object, quality control, finding and validating hits) and “Benchmark against MAGeCK” (1.15.11), which shows the simulation results and reruns the ScreenR methods on the K562 screen against precomputed MAGeCK results (`inst/extdata/benchmark`). Following the results of Section 6, the documentation of `find_gene_zscore_hit()` warns that non-targeting shRNAs are often not a good reference, and that of `find_roast_hit()` that ROAST and FRY need replicates in both groups.

4 Calibration of the p-values (1.15.10)

A first benchmark (Section 6.3) showed that the ScreenR tests were valid but much more conservative than their nominal level. Three causes were found and fixed.

Two-sided p-values used for a one-sided question. limma's `camera()` and `roast()` return two-sided p-values with the direction of the change; ScreenR selected the direction (for example "Down") and used the two-sided p-value, which is twice too large. Now

> one-sided p = p / 2 if the gene moves in the requested direction, 1 − p / 2 otherwise

and the FDR is computed on the one-sided p-values of all the genes.

A gene-level test with very few possible p-values. `find_gene_zscore_hit()` used a binomial test on the number of barcodes below a Z-score threshold; with 10 barcodes per gene it could only produce a handful of p-values, and the Benjamini-Hochberg procedure could rarely reach the nominal FDR. It is now a permutation test: the score of a gene with n barcodes (mean or median of their Z-scores) is compared with the scores of B random sets of n barcodes,

> p = (1 + number of random sets with score ≤ gene score) / (B + 1),

which is exact and has a resolution of $1/(B+1)$ (default $B = 10^5$). The random barcodes can be drawn from all the barcodes (default) or only from control genes (`control_genes` , e.g. non-targeting shRNAs), which removes the true hits from the reference.

No adaptation to the number of true hits. Benjamini-Hochberg controls the FDR at $\pi_0\alpha$, where π_0 is the fraction of genes without effect. `fdr_method = "storey"` estimates π_0 from the p-values above 0.5 and divides it out.

In addition `find_roast_hit(method = "fry")` uses `limma::fry()` , an analytical version of ROAST without the $1/(n_{rot}+1)$ floor of the rotation p-values, and the documentation now states that ROAST is a self-contained test (it asks "do the barcodes of this gene change?") while CAMERA is competitive ("do they change more than the other barcodes?").

5 Benchmark

5.1 Simulated screens

Each simulated screen has 500 genes with 10 shRNAs each and 500 non-targeting shRNAs. Library abundances are log-normal (sdlog 0.8); counts are negative binomial with mean proportional to abundance and depth (400 reads per shRNA, ±30% between samples). In the treated samples the shRNAs of the true hits (10% of the genes) have log2 fold change effect × e, where the knockdown efficiency e ~ Beta(4, 2) and 20% of the shRNAs are inactive (e = 0). Six scenarios were run with five seeds each:

Scenario	Change from the baseline
No hits	no true hits (calibration)
Baseline	3 vs 3 replicates, effect −1, dispersion 0.05
Weak effect	effect −0.5
2 replicates	2 vs 2 replicates
Noisy	dispersion 0.2
Off-target	3% of the shRNAs (targeting or not) depleted by −1.5 whatever their gene

Methods: MAGeCK 0.5.9.5 RRA (default median normalization), RRA with the non-targeting shRNAs as controls (`--control-sgrna`) and MLE (design with intercept and treatment, one-sided p-value from the Wald *z*, Benjamini-Hochberg FDR); ScreenR CAMERA (BH and Storey FDR), ROAST (999 rotations), FRY, the gene Z-score test with mean, median and control reference, the gene Z-score test on the edgeR statistics, and the legacy rule (`find_zscore_hit()` , 6 barcodes). For each method and nominal FDR level $\alpha \in \{1, 5, 10, 20\}$% we recorded the false discovery proportion (FDP), the true positive rate (TPR), the fraction of null genes with $p \leq \alpha$, the area under the precision-recall curve (AUPRC) and the run time.

5.2 Real screens

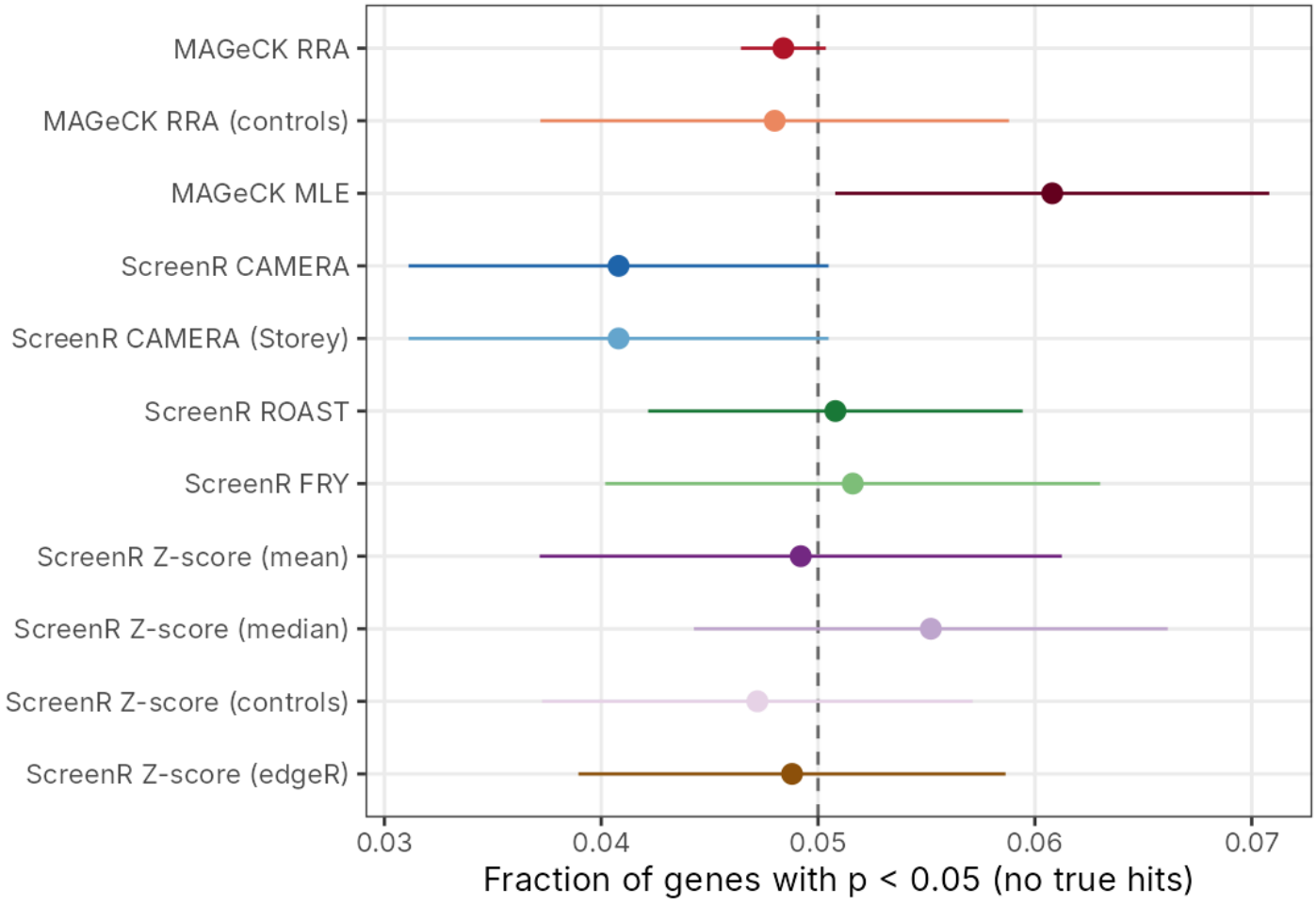
- **K562 essentiality screen** (dataset `k562_count_table`): plasmid vs day 14, depletion. Recall is measured on the core essential genes and false calls on the non-essential genes; the fraction of non-essential genes among the reference genes called estimates the FDR.
- **ScreenR example dataset** (`count_table`): day 4 treated vs control; without ground truth, used for the concordance between methods.

6 Results

6.1 Calibration with no true hits

Screens with no true hits: fraction of genes with p < 0.05 (should be 0.05) and mean number of genes called at 5% FDR (should be close to 0).

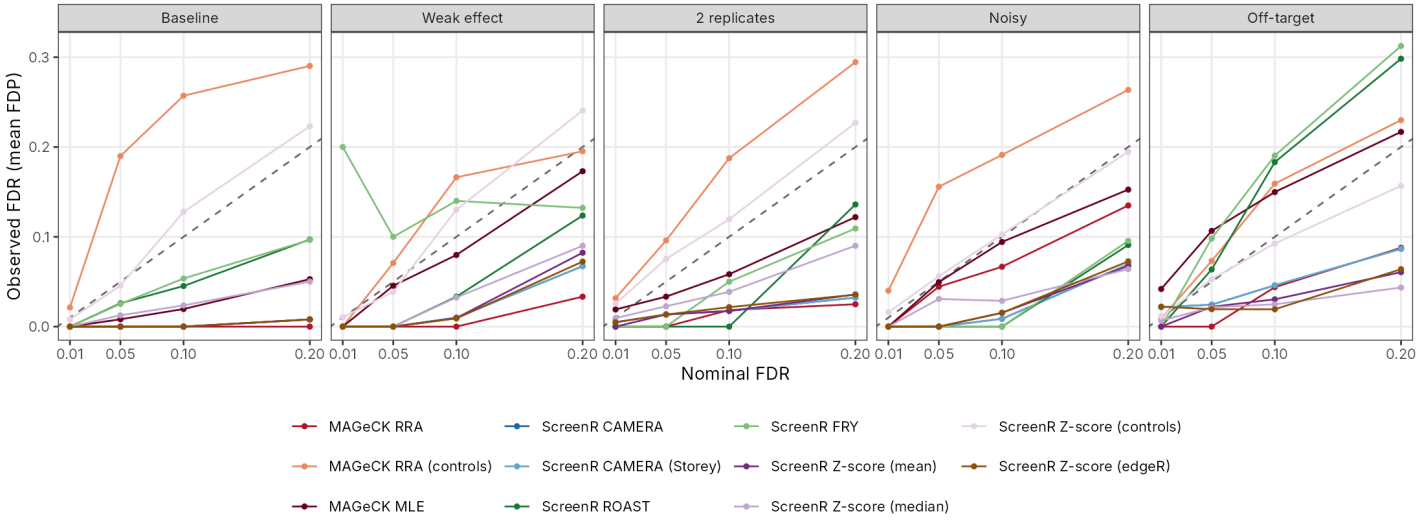
Method	Fraction p < 0.05	Genes called at 5% FDR
MAGeCK RRA	0.048	0.2
MAGeCK RRA (controls)	0.048	0.0
MAGeCK MLE	0.061	0.0
ScreenR CAMERA	0.041	0.0
ScreenR CAMERA (Storey)	0.041	0.0
ScreenR ROAST	0.051	0.0
ScreenR FRY	0.052	0.0
ScreenR Z-score (mean)	0.049	0.0
ScreenR Z-score (median)	0.055	0.0
ScreenR Z-score (controls)	0.047	0.0
ScreenR Z-score (edgeR)	0.049	0.0



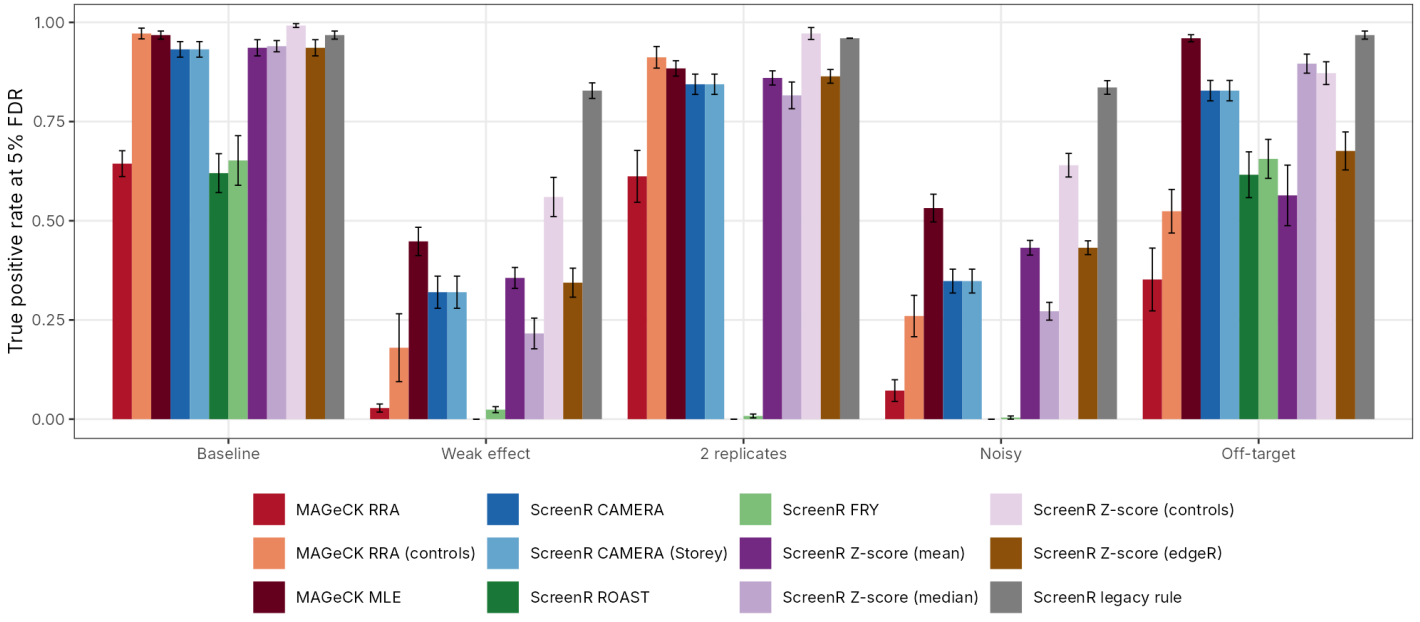
Fraction of genes with p < 0.05 in screens with no true hits (mean and 2 standard errors over seeds). The dashed line is the nominal 5%.

With no true hits every method had close to 5% of the genes with p < 0.05, so all the p-values are valid; MAGeCK MLE was slightly liberal (6.1%). At 5% FDR the calibrated methods called almost no gene, while the legacy ScreenR rule called on average 87 genes, all of them false.

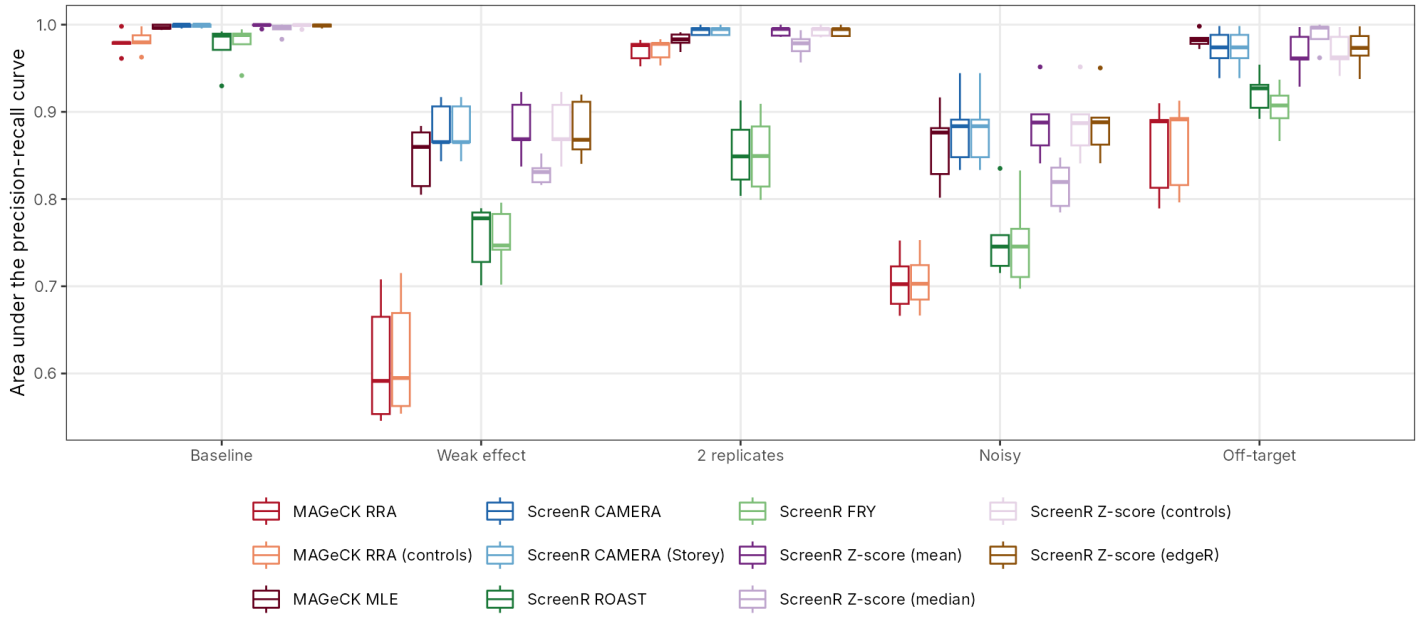
6.2 FDR control and power



Observed FDR (mean false discovery proportion over seeds) against the nominal FDR. Points on or below the dashed line control the FDR.



True positive rate at 5% FDR (mean and standard error over seeds).



Area under the precision-recall curve of the p-value ranking.

Summary at 5% FDR over the five scenarios with hits: mean FDP, worst scenario mean FDP, mean TPR, mean AUPRC and median run time (seconds).

Method	Mean FDP	Max FDP (scenario mean)	TPR	AUPRC	Time (s)
MAGeCK RRA	0.009	0.044	0.342	0.825	1.693
MAGeCK RRA (controls)	0.117	0.190	0.570	0.828	1.515
MAGeCK MLE	0.049	0.107	0.758	0.934	370.529
ScreenR CAMERA	0.008	0.024	0.654	0.945	1.676
ScreenR CAMERA (Storey)	0.008	0.024	0.654	0.945	1.500
ScreenR ROAST	0.018	0.064	0.247	0.852	3.431
ScreenR FRY	0.045	0.100	0.269	0.848	1.560
ScreenR Z-score (mean)	0.007	0.022	0.630	0.945	1.701
ScreenR Z-score (median)	0.018	0.031	0.628	0.921	6.852
ScreenR Z-score (controls)	0.054	0.075	0.807	0.946	0.339
ScreenR Z-score (edgeR)	0.007	0.019	0.650	0.946	1.660
ScreenR legacy rule	0.545	NA	0.912	NA	NA

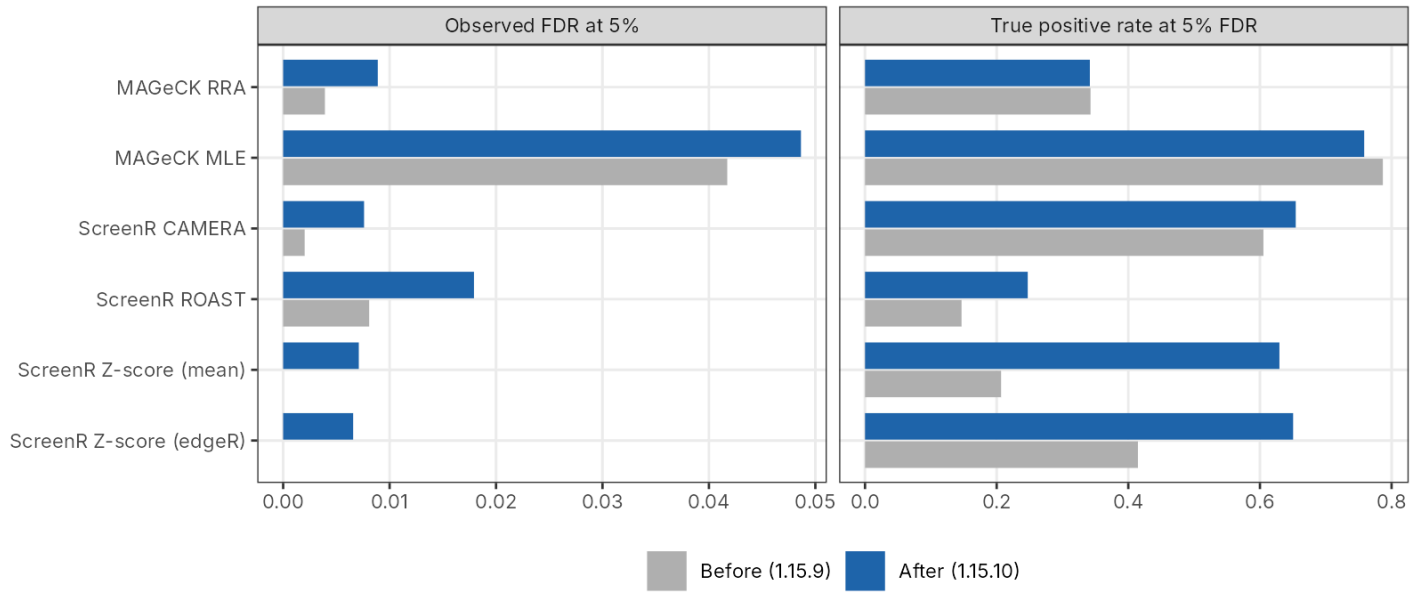
FDR control. CAMERA and the gene Z-score test (mean, median and on the edgeR statistics) kept the observed FDR below the nominal level in every scenario (worst scenario mean at 5%: 3.1%). They are still conservative when there are true hits, for two reasons. CAMERA is a competitive test and the Z-score test compares each gene with all the barcodes: the true hits are part of the reference and widen it. And Benjamini-Hochberg controls the FDR at $\pi_0\alpha$. The Storey correction did not change CAMERA: its one-sided p-values of the genes without effect are shifted towards 1 (they look “up” compared with an average lowered by the depleted hits), so the estimate of π_0 was above 1 (1.03–1.32) and was capped at 1.

MAGeCK MLE exceeded the nominal FDR with off-target effects (10.7% at 5%), and MAGeCK RRA with the non-targeting controls exceeded it in every scenario (up to 19%). The Z-score test with the non-targeting reference was close to nominal (3.9–7.5%). ROAST and FRY, which are self-contained tests, called shRNAs with off-target effects (FRY 9.8%, 10% in the weak-effect scenario). The legacy rule had an FDP of 50–60% in every scenario.

Power and ranking. In the baseline all the calibrated methods except ROAST, FRY and MAGeCK RRA found more than 90% of the hits. With weak effects or noisy data, at 5% FDR MAGeCK MLE found 45–53% of the hits, the ScreenR Z-score test 36–43%, CAMERA 32–35% and MAGeCK RRA 3–7%; the Z-score test with the non-targeting reference found the most (56–64%). The ranking of the genes (AUPRC), which does not depend on the FDR threshold, was best for CAMERA and the Z-score tests in four of the five scenarios (weak effect 0.88 vs 0.85 for MAGeCK MLE and 0.61 for MAGeCK RRA; noisy 0.89 vs 0.86 and 0.71); with off-target effects MAGeCK MLE (0.983) and the median Z-score (0.988) were best, since a median ignores single shRNAs with a seed effect. ROAST and FRY had little power with 2 replicates or noisy data: they test each gene on its own barcodes, so they rely on the replicate variance, while the competitive tests borrow information from all the genes.

Run time. MAGeCK MLE took a median of 371 s per screen; MAGeCK RRA and every ScreenR method 0.3–7 s (including the creation of the object).

6.3 Before and after the calibration



Observed FDR and true positive rate at 5% FDR before (1.15.9, first benchmark, 3 seeds) and after (1.15.10) the calibration fixes, mean over the scenarios with hits.

Before and after the calibration, mean over the scenarios with hits at 5% FDR. The two versions used different simulations (no non-targeting shRNAs and 3 seeds before), so the MAGeCK rows show the difference due to the simulation alone.

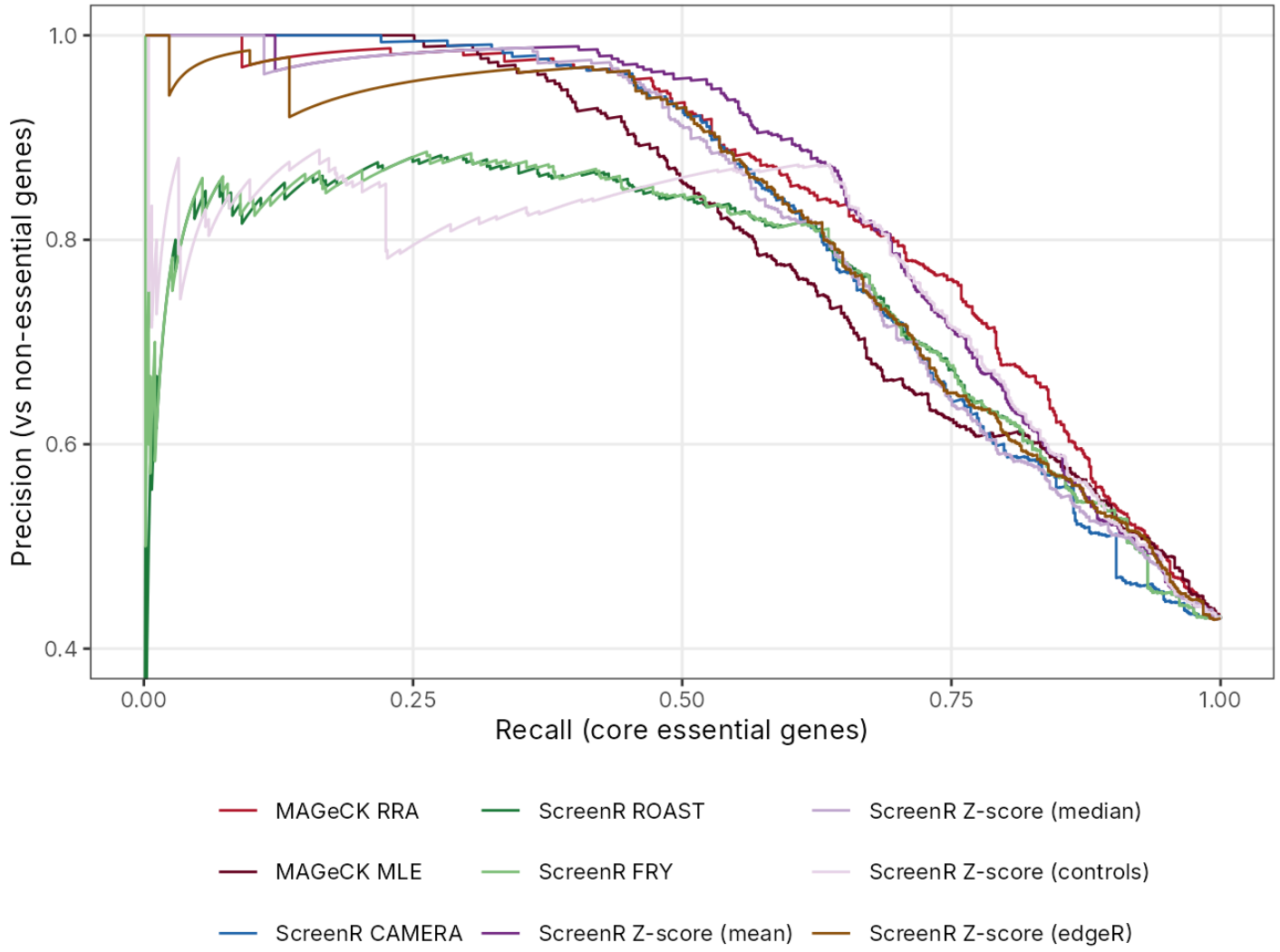
Method	version	FDP	TPR	AUPRC
MAGeCK RRA	Before (1.15.9)	0.004	0.343	0.864
MAGeCK RRA	After (1.15.10)	0.009	0.342	0.825
MAGeCK MLE	Before (1.15.9)	0.042	0.787	0.951
MAGeCK MLE	After (1.15.10)	0.049	0.758	0.934
ScreenR CAMERA	Before (1.15.9)	0.002	0.605	0.964
ScreenR CAMERA	After (1.15.10)	0.008	0.654	0.945
ScreenR ROAST	Before (1.15.9)	0.008	0.147	0.869
ScreenR ROAST	After (1.15.10)	0.018	0.247	0.852
ScreenR Z-score (mean)	Before (1.15.9)	0.000	0.207	0.887
ScreenR Z-score (mean)	After (1.15.10)	0.007	0.630	0.945
ScreenR Z-score (edgeR)	Before (1.15.9)	0.000	0.415	0.923
ScreenR Z-score (edgeR)	After (1.15.10)	0.007	0.650	0.946

The calibration fixes increased the power of every ScreenR method without losing FDR control: at 5% FDR the true positive rate of the Z-score test rose from 21% to 63%, that of the Z-score test on the edgeR statistics from 41% to 65%, CAMERA from 61% to 65% and ROAST from 15% to 25%, while the observed FDR stayed below 2%. The MAGeCK rows changed little, showing that the difference is due to the fixes and not to the new simulation.

6.4 K562 essentiality screen

K562: reference genes called at 5% FDR (681 core essential, 904 non-essential).

Method	Essential called	Non-essential called	Recall	Non-essential fraction	AUPRC	Time (s)
MAGeCK RRA	326	18	0.479	0.052	0.847	NA
MAGeCK MLE	417	130	0.612	0.238	0.809	NA
ScreenR CAMERA	312	17	0.458	0.052	0.824	7.4
ScreenR CAMERA (Storey)	312	17	0.458	0.052	0.824	7.6
ScreenR FRY	0	0	0.000	NA	0.757	7.2
ScreenR ROAST	0	0	0.000	NA	0.755	33.4
ScreenR Z-score (mean)	332	13	0.488	0.038	0.847	3.9
ScreenR Z-score (median)	328	25	0.482	0.071	0.821	7.3
ScreenR Z-score (controls)	547	292	0.803	0.348	0.767	0.8
ScreenR Z-score (edgeR)	348	32	0.511	0.084	0.817	2.7
ScreenR legacy rule	654	713	0.960	0.522	NA	NA



K562: precision-recall curves of the p-value ranking on the reference genes.

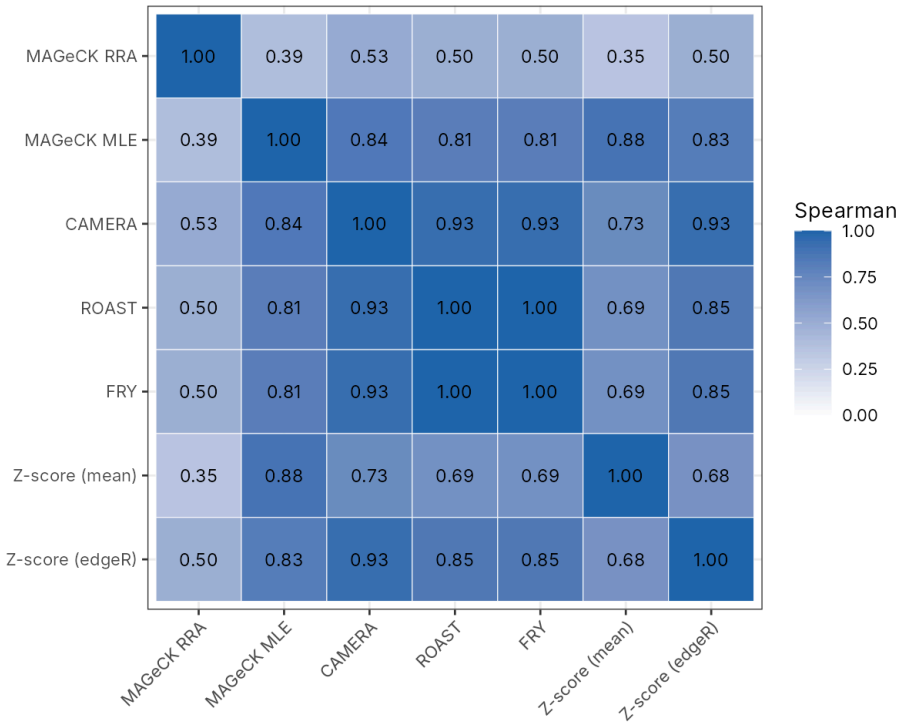
On the K562 screen the gene Z-score test (mean) found 332 of the 681 core essential genes at 5% FDR with 13 non-essential genes, MAGeCK RRA 326 with 18, and CAMERA 312 with 17; the ranking of the Z-score test was as good as MAGeCK RRA (AUPRC 0.847 vs 0.847) and MAGeCK MLE found 417 essential and 130 non-essential genes (AUPRC 0.809). The legacy rule called 654 essential but also 713 non-essential genes. MAGeCK MLE found the most essential genes but at the price of many false calls: 24% of the reference genes it called at 5% FDR were non-essential, in line with its liberal FDR in the simulations. MAGeCK MLE also skipped the genes with more than about 40 shRNAs (95 genes, 29 core essential and 17 non-essential), which are counted as not called; MAGeCK MLE was run with one permutation round (`--no-permutation-by-group --permutation-round 1`), since only its Wald statistic is used here and the default permutations would have taken more than ten hours.

Two results differ from the simulations. First, the Z-score test with the non-targeting shRNAs as reference called 292 non-essential genes: in shRNA screens nearly every targeting shRNA is somewhat depleted compared with non-targeting ones (off-target and RNAi-machinery toxicity), so the non-targeting shRNAs are not a good model of a gene without effect. Second, ROAST and FRY called no gene: with a single plasmid sample there is almost no replicate information for a self-contained test, although their ranking was still informative (AUPRC 0.76).

6.5 ScreenR example dataset

ScreenR example dataset, day 4 treated vs control: genes called at 5% FDR (the legacy rule has no FDR) and overlap with MAGeCK.

Method	Hits at 5% FDR	Shared with MAGeCK RRA	Shared with MAGeCK MLE
MAGeCK RRA	0	0	0
MAGeCK MLE	4	0	4
ScreenR CAMERA	0	0	0
ScreenR ROAST	9	0	4
ScreenR FRY	9	0	4
ScreenR Z-score (mean)	1	0	0
ScreenR Z-score (edgeR)	0	0	0
ScreenR legacy rule	97	0	4



ScreenR example dataset: Spearman correlation between the p-value rankings of the methods.

The ScreenR example dataset has a weak treatment effect: at 5% FDR MAGeCK RRA, CAMERA and the Z-score test on the edgeR statistics called no gene, MAGeCK MLE called 4 and ROAST and FRY 9, including the 4 of MAGeCK MLE. The legacy rule called 97 genes. The rankings of the ScreenR methods were highly correlated with MAGeCK MLE (Spearman 0.81–0.88) and less with MAGeCK RRA (0.35–0.53), which also agreed poorly with MAGeCK MLE (0.39).

7 Discussion

The legacy rule should not be used to call hits. In every simulated scenario about half of its hits were false, it called about 87 genes in screens without any effect and half of its K562 calls were non-essential genes. It is kept for compatibility, but the vignettes now use the calibrated tests.

Recommended methods. The gene Z-score permutation test (mean) and CAMERA control the FDR in all the scenarios, rank the hits as well as or better than MAGeCK MLE, are as good as MAGeCK RRA on the K562 screen and run in seconds, 100–300 times faster than MAGeCK MLE. At a fixed FDR they are more conservative than MAGeCK MLE, which finds more hits with weak or noisy effects but exceeded the nominal FDR with off-target effects. The median Z-score is the most robust to off-target shRNAs. ROAST and FRY answer a different question (does the gene change at all?) and need replicates in both conditions.

Controls. A reference of non-targeting shRNAs increased the power when they behave like genes without effect (simulation), but produced many false hits in the K562 screen, where they do not. Non-targeting shRNAs should be used as reference only after checking, for example, that their fold changes match those of known non-essential genes; the default reference (all the barcodes) is safer.

Limitations and future work. The simulations use a simple model of knockdown efficiency and off-target effects; the K562 evaluation is limited to an essentiality screen with one plasmid sample. The conservativeness of the competitive tests when many genes are hits could be reduced by an empirical-null or iterative reference (removing the strongest hits from the reference), and the power with weak effects by weighting the shRNAs by their estimated efficiency, as MAGeCK MLE does. A larger evaluation on several screens with known hits (for example drug-resistance screens with validated genes) would strengthen the comparison.

8 Reproducibility

All the code is in the `benchmark/` folder of the ScreenR repository (branch `devel`), excluded from the package build:

- `install_mageck.sh` installs MAGeCK 0.5.9.5;
- `simulation/` simulates the screens and runs the methods (`run_benchmark.R`), checks the null calibration (`calibration_null.R`) and keeps the first benchmark (`v1_before_calibration/`);
- `realdata/` runs the K562 and ScreenR-dataset comparisons;
- `summarise_results.R` produces every table and figure of this report from `results/`, so the report can be rebuilt without rerunning MAGeCK;
- `paper/ScreenR_report.Rmd` is this report (rendered to HTML and printed to `paper/ScreenR_report.pdf`).

The K562 dataset is built by `inst/scripts/k562_essential_screen.R`. The package passes `R CMD check` with 219 tests.

9 References

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